CLASS 9 · CBSE · GANITA MANJARI PART 1 · NCERT SOLUTIONS

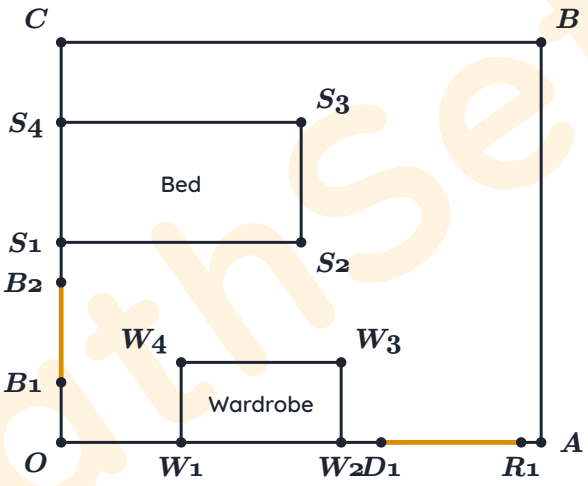
Chapter 1: Orienting Yourself: The Use of Coordinates

Step-by-step solutions to the textbook exercises: 26 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise Set 1.1

Fig. These questions use the plan of a bedroom drawn on coordinate axes, with the corner O as the origin and 1 unit $=$ 1 ft. The plan is redrawn below with the points that are needed.



O(0, 0), A(12, 0), B(12, 10), C(0, 10), D1(8, 0), R1(11.5, 0), B1(0, 1.5), B2(0, 4)

SOLUTION

Read each point from the plan: the first number is the distance along the x-axis and the second is the distance along the y-axis.

(i) D_1R_1 is the door of the room. How far is the door from the left wall (the y-axis), and how far is it from the x-axis?

SOLUTION

The door starts at D_1, which is 8 units to the right of O. So the door is 8 ft from the left wall.

Both D_1 and R_1 lie on the x-axis, so the distance of the door from the x-axis is 0.

Answer: 8 ft from the y-axis and 0 ft from the x-axis

(ii) Write the coordinates of D_1.

SOLUTION

D_1 is on the x-axis, 8 units from O, so its y-coordinate is 0.

Answer: $D_1 = (8, 0)$

(iii) R_1 is the point $(11.5, 0)$. How wide is the door? Is this width comfortable, and can a person in a wheelchair enter easily?

SOLUTION

Both points are on the x-axis, so the width is the difference of the x-coordinates: $11.5 - 8 = 3.5$ ft.

This is a comfortable width for a room door. A wheelchair is a little over 2 ft wide, so it can pass through a 3.5 ft door easily.

Answer: 3.5 ft, which is wide enough for a wheelchair

(iv) $B_1(0, 1.5)$ and $B_2(0, 4)$ are the ends of the bathroom door. Is it narrower or wider than the room door?

SOLUTION

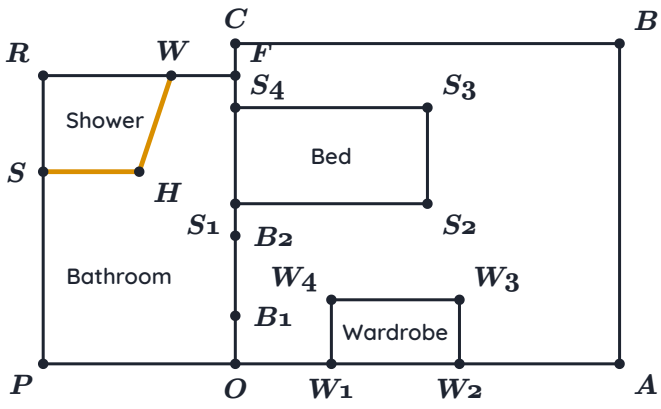
Both points are on the y-axis, so the width is the difference of the y-coordinates: $4 - 1.5 = 2.5$ ft.

$2.5 < 3.5$, so the bathroom door is narrower than the room door by 1 ft.

Answer: narrower (2.5 ft against 3.5 ft)

2 Exercise Set 1.2

Fig. The plan now also shows the bathroom to the left of the y-axis. It is redrawn below. Reading from the plan: $P(-6, 0)$, $R(-6, 9)$, $F(0, 9)$, $S(-6, 6)$, $H(-3, 6)$, $W(-2, 9)$, the wardrobe corners $W_1(3, 0)$, $W_2(7, 0)$, $W_3(7, 2)$, $W_4(3, 2)$ and the bed corners $S_1(0, 5)$, $S_2(6, 5)$, $S_3(6, 8)$, $S_4(0, 8)$.



The bedroom and the bathroom, with 1 unit = 1 ft

SOLUTION

Points to the left of the y-axis have a negative x-coordinate.

Q1 A rectangular study table has three of its feet at $(8, 9)$, $(11, 9)$ and $(11, 7)$. (i) Where is the fourth foot? (ii) Is this a good spot for the table? (iii) What are the width and the length of the table? Can its height be found?

SOLUTION

(i) The feet form a rectangle with sides parallel to the axes. The x-coordinates used are 8 and 11 and the y-coordinates are 9 and 7. The missing combination is $(8, 7)$.

(ii) The table lies between $x = 8$ and $x = 11$ and between $y = 7$ and $y = 9$. It is clear of the bed (which ends at $x = 6$), of the wardrobe and of the door on the bottom wall, and it is close to the corner B. So it is a sensible spot. Other answers are possible if you give a reason.

(iii) Length $= 11 - 8 = 3$ ft and width $= 9 - 7 = 2$ ft. The height cannot be found, because the plan shows the room from above and has no third coordinate.

Answer: (i) $(8, 7)$ (ii) yes (iii) 3 ft long and 2 ft wide; the height cannot be found

Q2 The bathroom door B_1B_2 has its hinge at B_1 and opens into the bedroom. Will it hit the wardrobe? What would you change if the door were made wider?

SOLUTION

The door is $4 - 1.5 = 2.5$ ft wide. As it opens, its free end moves on a circle of radius 2.5 ft with centre $B_1(0, 1.5)$.

The nearest point of the wardrobe to B_1 is $(3, 1.5)$ on the side W_1W_4, which is 3 ft away. Since $2.5 < 3$, the door does not reach the wardrobe.

A door of width 3 ft or more would hit it. Then the door could be hinged at B_2, made to open into the bathroom, or the wardrobe could be moved to the right.

Answer: no, the door stops 0.5 ft short of the wardrobe

Q3 (i) Write the coordinates of the corners O, F, R and P of the bathroom. (ii) What shape is the showering area $SHWR$? Write the coordinates of its corners. (iii) Mark a 3 ft $\times$ 2 ft space for a washbasin and a 2 ft $\times$ 3 ft space for a toilet, and write the coordinates of their corners.

SOLUTION

(i) $O(0, 0)$, $F(0, 9)$, $R(-6, 9)$ and $P(-6, 0)$.

(ii) $S(-6, 6)$, $H(-3, 6)$, $W(-2, 9)$ and $R(-6, 9)$. The sides SH and RW are both parallel to the x-axis but have different lengths (3 ft and 4 ft), so $SHWR$ is a trapezium.

(iii) One possible answer. Washbasin in the corner P: $(-6, 0)$, $(-3, 0)$, $(-3, 2)$, $(-6, 2)$. Toilet against the left wall: $(-6, 3)$, $(-4, 3)$, $(-4, 6)$, $(-6, 6)$. Any spaces of the right size that stay inside the bathroom and do not overlap are correct.

Answer: (i) as above (ii) a trapezium (iii) answers differ

Q4 (i) The room door opens from a dining room 18 ft long and 15 ft wide, whose length runs from P to A. Sketch it and write the coordinates of its corners. (ii) A 5 ft $\times$ 3 ft dining table is placed exactly in the centre of the dining room. Write the coordinates of its feet.

SOLUTION

(i) PA lies on the x-axis from $x = -6$ to $x = 12$, a length of 18 ft. The door is on this wall, so the dining room lies below the x-axis and reaches down to $y = -15$.

Its corners are $P(-6, 0)$, $A(12, 0)$, $(12, -15)$ and $(-6, -15)$.

(ii) The centre of the dining room is $\left(\frac{-6+12}{2}, \frac{0-15}{2}\right) = (3, -7.5)$.

With the 5 ft side along the length of the room, the feet are 2.5 ft to the left and right of the centre and 1.5 ft above and below it: $(0.5, -9)$, $(5.5, -9)$, $(5.5, -6)$ and $(0.5, -6)$.

Answer: (i) $(-6, 0)$, $(12, 0)$, $(12, -15)$, $(-6, -15)$ **(ii)** $(0.5, -9)$, $(5.5, -9)$, $(5.5, -6)$, $(0.5, -6)$

3 End-of-Chapter Exercises

Q1 What are the x-coordinate and the y-coordinate of the point where the two axes meet?

SOLUTION

The axes meet at the origin, which is at zero distance from both axes.

Answer: both are 0, so the point is $(0, 0)$

Q2 The point W has x-coordinate -5. What can you say about the coordinates of a point H on the line through W parallel to the y-axis? In which quadrants can H lie?

SOLUTION

Every point on a line parallel to the y-axis has the same x-coordinate. So $H = (-5, y)$ for some number y.

If $y > 0$, H is in Quadrant II. If $y < 0$, H is in Quadrant III. If $y = 0$, H is on the x-axis.

Answer: $H = (-5, y)$**, in Quadrant II or Quadrant III**

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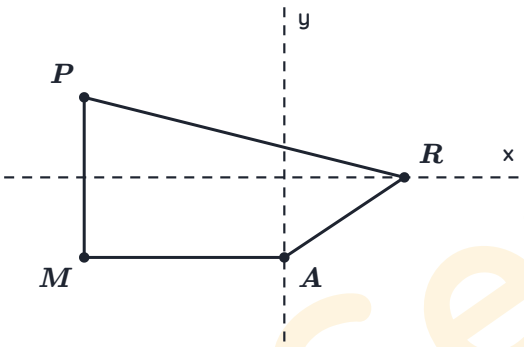
Q3 The points $R(3, 0)$, $A(0, -2)$, $M(-5, -2)$ and $P(-5, 2)$ are joined in order. Predict (i) two sides of $RAMP$ that are perpendicular (ii) a side that is parallel to an axis (iii) two points that are mirror images in an axis, and name the axis.

SOLUTION

(i) A and M have the same y-coordinate, so AM is horizontal. M and P have the same x-coordinate, so MP is vertical. So $AM \perp MP$.

(ii) AM is parallel to the x-axis (and MP is parallel to the y-axis).

(iii) $M(-5, -2)$ and $P(-5, 2)$ differ only in the sign of the y-coordinate, so they are mirror images in the x-axis.



Answer: (i) AM and MP (ii) AM (iii) M and P, in the x-axis

Q4 Plot $Z(5, -6)$. Construct a right-angled triangle IZN and find the lengths of its three sides.

SOLUTION

One possible answer: take $N(1, -6)$ and $I(5, -3)$. Then ZN is horizontal and ZI is vertical, so the right angle is at Z.

$ZN = 5 - 1 = 4$ units and $ZI = -3 - (-6) = 3$ units. By the Baudhayana-Pythagoras theorem, $IN = \sqrt{4^2 + 3^2} = 5$ units.

Answer: sides 3, 4 and 5 units for this choice. Other triangles are also correct.

Q5 What would a system of coordinates be like without negative numbers? Could it locate every point of the plane?

SOLUTION

With only zero and positive numbers, we could give coordinates only to the origin, the positive halves of the two axes and the points of Quadrant I.

Points to the left of the y-axis or below the x-axis could not be named, so the system would not locate all the points of the plane.

Answer: only Quadrant I and its edges could be located, so not every point

Q6* Are $M(-3,-4)$, $A(0,0)$ and $G(6,8)$ on one straight line? Give a method that does not need plotting.

SOLUTION

Method: find the three distances. Three points lie on one line exactly when the two shorter distances add up to the longest.

$MA = \sqrt{3^2+4^2} = 5$, $AG = \sqrt{6^2+8^2} = 10$ and $MG = \sqrt{9^2+12^2} = \sqrt{225} = 15$.

$MA + AG = 5 + 10 = 15 = MG$.

Answer: yes, the three points are on one straight line

Q7* Use the same method to check whether $R(-5,-1)$, $B(-2,-5)$ and $C(4,-12)$ are on one straight line.

SOLUTION

$RB = \sqrt{3^2+4^2} = 5$, $BC = \sqrt{6^2+7^2} = \sqrt{85} \approx 9.22$ and $RC = \sqrt{9^2+11^2} = \sqrt{202} \approx 14.21$.

$RB + BC \approx 14.22$, which is close to RC but not equal to it. To be sure: if $5 + \sqrt{85} = \sqrt{202}$, squaring gives $110 + 10\sqrt{85} = 202$, so $\sqrt{85} = 9.2$ and $85 = 84.64$, which is false.

Answer: no. The points are nearly, but not exactly, in a line, which a plotted graph may not show clearly.

Q8* Take the origin as one vertex and give the vertices of (i) a right-angled isosceles triangle (ii) an isosceles triangle whose other two vertices lie in Quadrant III and Quadrant IV.

SOLUTION

(i) One possible answer: $O(0,0)$, $(4,0)$ and $(0,4)$. The two sides along the axes are 4 units each and meet at a right angle at O.

(ii) One possible answer: $O(0,0)$, $(-3,-4)$ and $(3,-4)$. Each of the first two sides is $\sqrt{3^2+4^2} = 5$ units, so the triangle is isosceles.

Answer: (i) $(0,0)$, $(4,0)$, $(0,4)$ **(ii)** $(0,0)$, $(-3,-4)$, $(3,-4)$. **Other answers are possible.**

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Q9* For each row, say whether M is the mid-point of ST, with a reason. What connects the coordinates of M, S and T when M is the mid-point?

S	M	T
$(-3, 0)$	$(0, 0)$	$(3, 0)$
$(2, 3)$	$(3, 4)$	$(4, 5)$
$(0, 0)$	$(0, 5)$	$(0, -10)$
$(-8, 7)$	$(0, -2)$	$(6, -3)$

SOLUTION

M is the mid-point when it is on the segment and $SM = MT$.

S, M, T	Mid-point?	Reason
$(-3, 0), (0, 0), (3, 0)$	Yes	All on the x-axis, and $SM = MT = 3$
$(2, 3), (3, 4), (4, 5)$	Yes	$SM = MT = \sqrt{2}$ and $ST = 2\sqrt{2}$
$(0, 0), (0, 5), (0, -10)$	No	M is above S but T is below it, so M is not between them
$(-8, 7), (0, -2), (6, -3)$	No	$SM = \sqrt{145}$ but $MT = \sqrt{37}$

In the two "Yes" rows, each coordinate of M is the average of the matching coordinates of S and T: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Answer: Yes, Yes, No, No. The mid-point has the averages of the coordinates.

Q10* $M(-7, 1)$ is the mid-point of $A(3, -4)$ and $B(x, y)$. Find B.

SOLUTION

$\frac{3 + x}{2} = -7$ gives $x = -14 - 3 = -17$.

$\frac{-4 + y}{2} = 1$ gives $y = 2 + 4 = 6$.

Answer: $B = (-17, 6)$

Q11* P and Q trisect AB, with P nearer to A. Using mid-points, find P and Q when $A = (4, 7)$ and $B = (16, -2)$.

SOLUTION

Since $AP = PQ = QB$, P is the mid-point of AQ and Q is the mid-point of PB. Let $P = (a, b)$ and $Q = (c, d)$.

x-coordinates: $2a = 4 + c$ and $2c = a + 16$. Putting $c = 2a - 4$ in the second: $4a - 8 = a + 16$, so $a = 8$ and $c = 12$.

y-coordinates: $2b = 7 + d$ and $2d = b - 2$. Putting $d = 2b - 7$ in the second: $4b - 14 = b - 2$, so $b = 4$ and $d = 1$.

Answer: $P = (8, 4)$ **and** $Q = (12, 1)$

Q12* (i) Show that $A(1,-8)$, $B(-4,7)$ and $C(-7,-4)$ lie on a circle K with centre $O(0,0)$, and find its radius. (ii) Are $D(-5,6)$ and $E(0,9)$ inside, on or outside K?

SOLUTION

(i) $OA=\sqrt{1+64}=\sqrt{65}$, $OB=\sqrt{16+49}=\sqrt{65}$ and $OC=\sqrt{49+16}=\sqrt{65}$. The three points are at the same distance from O, so they lie on a circle with centre O and radius $\sqrt{65}\approx 8.06$ units.

(ii) $OD=\sqrt{25+36}=\sqrt{61}<\sqrt{65}$, so D is inside the circle. $OE=9=\sqrt{81}>\sqrt{65}$, so E is outside the circle.

Answer: (i) radius $\sqrt{65}$ units (ii) D is inside and E is outside

Q13* The mid-points of the sides of $\triangle ABC$ are $D(5,1)$, $E(6,5)$ and $F(0,3)$. Find A, B and C.

SOLUTION

Take D, E and F as the mid-points of BC, CA and AB. Adding the coordinates in pairs: $B+C=2D=(10,2)$, $C+A=2E=(12,10)$ and $A+B=2F=(0,6)$.

Adding all three, $2(A+B+C)=(22,18)$, so $A+B+C=(11,9)$.

$A=(11,9)-(10,2)=(1,7)$, $B=(11,9)-(12,10)=(-1,-1)$ and $C=(11,9)-(0,6)=(11,3)$.

Answer: $A(1,7)$, $B(-1,-1)$, $C(11,3)$

Q14 Two main roads of a city cross at its centre, one running North-South and the other East-West. All other streets are parallel to these and 200 m apart, with 10 streets in each direction. (i) Draw a model with 1 cm $=$ 200 m. (ii) The crossing of the 2nd North-South street and the 5th East-West street is called $(2,5)$. How many crossings can be called (a) $(4,3)$ (b) $(3,4)$?

SOLUTION

(i) Draw 10 vertical lines 1 cm apart for the North-South streets and 10 horizontal lines 1 cm apart for the East-West streets, and number each set from 1 to 10.

(ii) Two streets in different directions cross at exactly one point. So only one crossing is $(4,3)$: the 4th North-South street with the 3rd East-West street. In the same way only one crossing is $(3,4)$.

These are two different crossings, because the order of the two numbers matters.

Answer: (a) one (b) one

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Q15 A screen is 800 pixels wide and 600 pixels high, with the origin at its bottom-left corner. One circular icon has centre $A(100, 150)$ and radius 80. Another has centre $B(250, 230)$ and radius 100. (i) Does any part of either circle lie outside the screen? (ii) Do the circles intersect?

SOLUTION

(i) The first circle runs from $x = 20$ to $x = 180$ and from $y = 70$ to $y = 230$. The second runs from $x = 150$ to $x = 350$ and from $y = 130$ to $y = 330$. All of these are between 0 and 800 for x and between 0 and 600 for y, so both circles are fully on the screen.

(ii) $AB = \sqrt{150^2 + 80^2} = \sqrt{28900} = 170$. The sum of the radii is $80 + 100 = 180$ and their difference is 20.

Since $20 < 170 < 180$, the circles cut each other at two points.

Answer: (i) no (ii) yes, they intersect

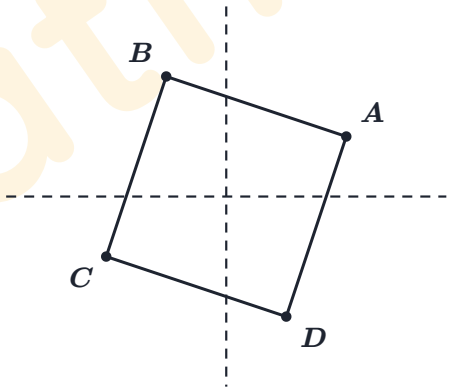
Q16 Plot $A(2, 1)$, $B(-1, 2)$, $C(-2, -1)$ and $D(1, -2)$. Is $ABCD$ a square? Why? Find its area.

SOLUTION

Sides: $AB = \sqrt{3^2 + 1^2} = \sqrt{10}$, $BC = \sqrt{1^2 + 3^2} = \sqrt{10}$, $CD = \sqrt{10}$ and $DA = \sqrt{10}$. All four sides are equal.

Diagonals: $AC = \sqrt{4^2 + 2^2} = \sqrt{20}$ and $BD = \sqrt{2^2 + 4^2} = \sqrt{20}$. The diagonals are equal too.

A quadrilateral with four equal sides and equal diagonals is a square. Its area is $(\sqrt{10})^2 = 10$ square units.



Answer: yes, $ABCD$ is a square of area 10 square units

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